



TYPES OF TRANSPORT

Logistics transport is broken down into one of four modes, namely road, air, ocean and rail freight.

Sea freight:

Before the 1950's most ocean freight was carried onboard bulk vessels. This involved a process of manually loading items and similar to how a jig saw puzzle is constructed, cargo was packaged accordingly. In 1956 an entrepreneur by the name of Malcom McLean put 58 aluminium truck bodies aboard an ageing tanker ship (called the Ideal-X) which set sail from Newark, NJ to Houston, TX in the United States.

This marked the start of containerised transport as we know it today.² Containers can be stacked on top of each other onboard the ship, thus allowing very efficient space utilisation and cargo handling. Furthermore, freight could now move from origin to destination across many modes and services with greater ease of handling. The introduction and growth of containerisation led to huge changes in ports which previously were dominated by large workforces responsible for manual handling of bulk cargo. Containerisation also reduced the costs of transporting freight by maritime transport and significantly improved its efficiency. Containerisation spread to other modes and various alliances were formed between combinations of transport companies.

Types of Containers:

Costings:

- Ocean Freight is always quoted in USD
- A Freight Ton is equal to 1000 Kg or 1 CBM
- For cargo that doesn't need to be transported as a full container load (FCL) the option of a consolidated load exist. This is known as the less than container load (LCL)



- Thus the freight for a LCL consignment will be calculated based on whichever is the highest between the Kg or CBM of the consignment
- In the case of an FCL consignment, the freight will be charged per container e.g. USD 4500
- A Break Bulk consignment will be calculated in Harbour Ton

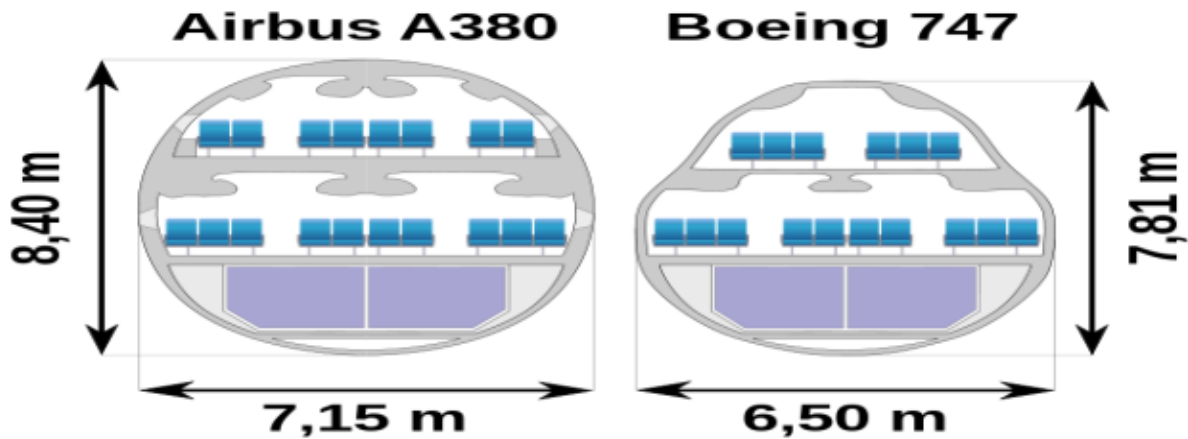
Airfreight:

Generally the most expensive means of transporting cargo but ultimately the quickest means of transportation.

Air transportation is broken down into two distinct aircraft types, namely narrow body and widebody.

Widebodies

- These are aircraft that twin isled
- These are aircraft types such as the Airbus family of A380, A330, A340 and the Boeing family such as the 747, 777 and cargo aircraft such as the 747F
- Max dimensions are 300 x 200 x 155cm – Passenger aircraft
- Max dimensions are 600 x 200 x 240cm – cargo aircraft
- Max weight per piece is 3500kg for passenger aircraft and 10 000kgs for cargo aircraft
- Due to the recent covid pandemic a number of formally passenger aircraft have undergone retro fits which have enabled them to be converted to handle certain types of cargo within the cabin. Many are now being returned back to fully passenger aircraft as passenger volumes increase



Narrow bodies

- Defined by single isles
- Include the workhorse of regional travel in the B737 and 737F as well as the smaller Airbus and Embraer aircraft types
- These are typically used for regional flights and typical used by low cost airlines for quick turnarounds
- Cargo restricted to 100 x 80 x 80cm, 120kgs max per piece

Airfreight is moved on a unit loading device (ULD)

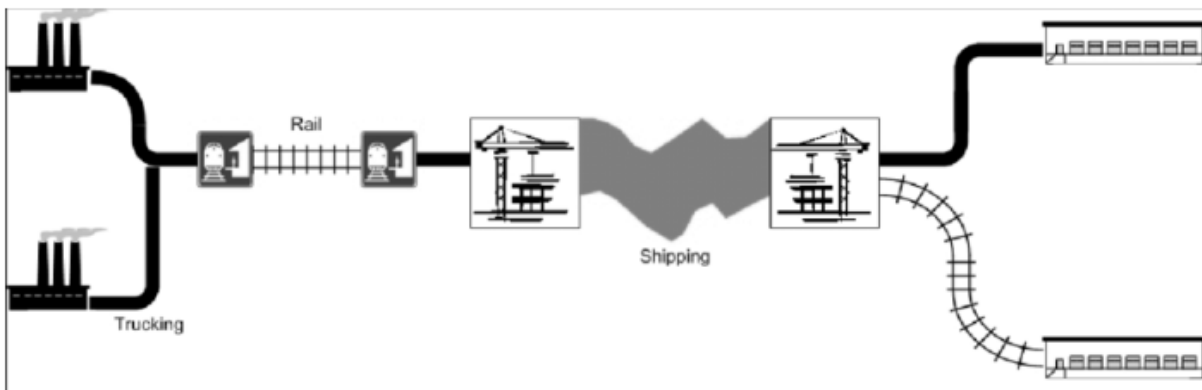
- A ULD is a pallet that passenger bags or cargo is loaded into
- This results in quicker loading and offloading which improves the efficiency of the handling agent
- A ULD has a manifest that accompanies it so that its content can be tracked

Air freight Calculation:

- Airfreight is charged on either actual KG or volumetric weight
- As an example a box of motor vehicle parts has an actual weight of 250 kg. the dimensions of the box are 120 cm X 130 cm X 210 cm = 3276000 divided by 6000 = 546 kg
- Thus the volumetric weight is higher than the actual weight meaning the volumetric will be charged by the airline

Road Freight:

Road freight is typically used to move cargo over shorter destinations and can be used as part of an intermodal transport offering of ocean, road, rail and air. As per the below road can be utilised to get goods to a rail head or a final destination.



An intermodal transportation network

Alternatively road in much of Southern Africa is the only means of getting cargo to landlocked countries as the movement of cargo via rail is slow.



Road freight can be based on a dedicated load or a consolidated load which is to a depot in the respective country of destination

Rail Freight

Rail freight is commonly used as part of the intermodal service offering to bring containers from port to the inland port. In South Africa, Johannesburg has City Deep and Pretoria has Pretcon as the respective inland container depots. In addition to this Durban to Johannesburg rail line is one of the busiest rail lines in South Africa. From City Deep containers are sent either specific industrial sidings such as those owned by BMW and Nissan. Alternatively trains are made up and containers are sent over border to Botswana, Zimbabwe and Zambia.

As a colonial legacy much of the network of Southern Africa is constructed to take minerals and produce from the interior to the ports for export purposes. The South African network from a freight perspective consist of four systems, namely:

1. Iron ore and manganese system
2. Coal system
3. North-eastern system
4. Intermodal and general freight system.

These systems are run independently with their own procurement, stock and general managers which as individual business units report through to the head office Johannesburg.